1. Title Page

Project Title: Temperature Forecasting Using Statistical, Machine Learning, and Deep Learning Time Series Models

Student: Lilit Poghosyan

Course: Time Series Analysis

Instructor: Sachin Kumanr

Submission Date: 05/14/2026

2. Abstract

This project aims at forecasting hourly temperature values using multiple time series forecasting methods. This is a historical weather data set from 2006-2016, with high level of seasonality and cyclicality. The primary goal of the project was to evaluate and benchmark the performance of classical statistical forecasting models against machine learning and deep learning models to identify the optimal model for forecasting nonlinear and seasonal weather data.

Models used are ARIMA, SARIMA, Exponential Smoothing, VAR, Random Forest, Gradient Boosting, SVR, LSTM, GRU, CNN. To start, exploratory time series analysis was conducted by visualizing the data, performing autocorrelation analysis, decomposing the data into trend, seasonal, and irregular components, and testing the stationarity of the data using the Augmented Dickey-Fuller (ADF) test.

The results indicated that the statistical models had difficulties to capture the complex non-linear structure of the temperature series, particularly in the longer forecasting horizons. The performance of the machine learning and deep learning models was much better. The Gradient Boosting technique was the most successful method with an RMSE of about 0.77 and an R^2 value of 0.993. The other models, such as LSTM, GRU, CNN, and Random Forest also gave very good predictions.

In general, this project proves the efficiency of the advanced machine learning and deep learning techniques for weather forecasting problems with strong seasonality and nonlinear temporal patterns.

3. Introduction

In finance, energy systems, economics and weather forecasting, among other areas, the time series forecasting is a very common problem. The significance of temperature forecasting is that it plays a critical role in supporting farm, road, and energy planning, as well as climate-related decision making.

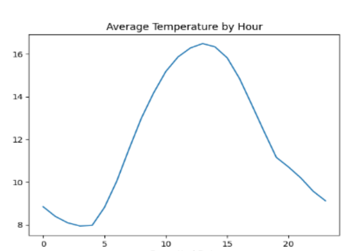
This project is aimed at analysing the historical hourly temperature data and predicting temperature values in the future with various forecasting methods. The aim of the project is to do the comparison between traditional statistical and modern machine learning or deep learning techniques in order to test their capacity to capture the trend, seasonality and nonlinear temporal behavior.

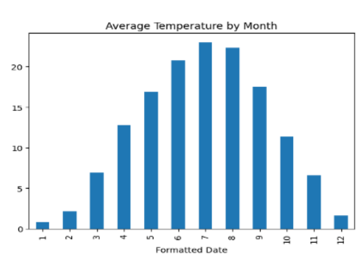
The project in particular explores the performance of the various models to predict highly seasonal weather. Particular focus is placed on the ability of models to deal with the repetition of daily and yearly cycles, the occurrence of abrupt changes and long-term dependencies.

4. Dataset Description

Data for this project was obtained from the hourly weather observations recorded from 2006 to 2016. Temperature (in degrees Celsius) was the target variable for forecasting. The data base is predominantly used as a univariate time series since the forecasting task is to predict the future values of temperature. Some machine learning models also employed engineered lag features and rolling statistics from the original series. The dataset has the following key features: hourly frequency, strong yearly seasonality, daily seasonal cycles, nonlinear fluctuations, significant seasonal variations in temperatures.

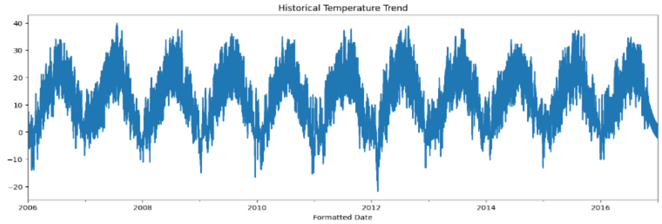
This data indicated distinct warm and cold periods in various months of the year. The mean temperature for the months of the year were of the order of 23°C in summer and near 0°C in winter.



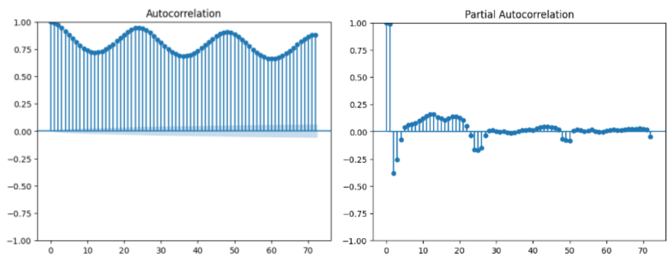


5. Exploratory Time Series Analysis

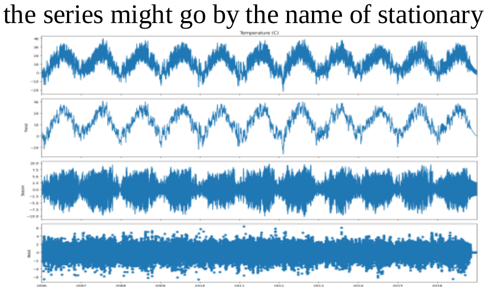
Initial exploratory analysis showed very interesting repeating seasonal patterns in the data. The average temperature was clearly seasonal throughout the day, with higher temperatures during the day and lower temperatures at night. The annual time series plot also showed good annual seasonality in the form of a summer/winter cycle.



Strong temporal dependence was found in observations as indicated by autocorrelation analysis. The ACF plot revealed that the correlation was decreasing and increasing gradually which shows high seasonality and persistence in the series. Likewise, the PACF plot showed some large spikes at lag 2 or lag 3, indicating the presence of AR behavior.



The Augmented Dickey-Fuller (ADF) test was used to evaluate stationarity. The test produced ADF Statistic: -7.399 and P-value: 0.000000001. As p value was < 0.05, the null hypothesis of non-stationarity was rejected. This means that the series might go by the name of stationary without having to differ it heavily.

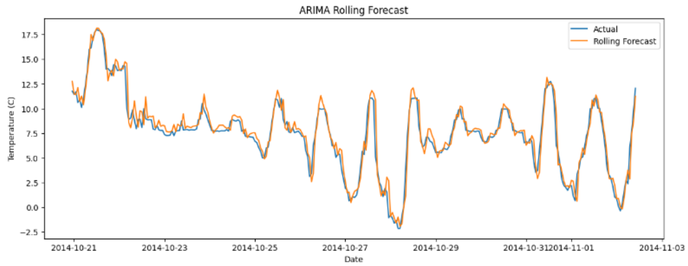


Decomposition using STL was also applied to decompose the series into the trend, seasonal and residual components. It was evident that there was strong seasonal structure and random noise in the decomposition.

6. Methodology

During the project, an array of forecast techniques were adopted and contrasted. The statistical forecasting techniques used were: ARIMA, SARIMA, Exponential Smoothing, VAR. These models were chosen because they are used widely in traditional time series forecasting and are able to model explicitly the trend and the seasonality.

For ARIMA, the rolling forecasts were also applied to make the forecasts more realistic by continually updating the model with the newly observed values.



Random Forest, Gradient Boosting, Support Vector Regression (SVR) were the machine learning techniques used. The following models incorporated engineered lag and rolling statistics. The models were given a series of lagged variables, which enabled them to learn how the past temperatures affect observations of the future. Engineered features were examples of: Previous hourly temp, Rolling average (rolling_24)), and Seasonal lag variables (lag_1, lag_24). GridSearchCV was used to tune the hyperparameters.

Deep Learning Models were trained directly from sequential observations in place of conventional manually-designed lag variables. A window of 24 hours was used for lookback, which implies that every prediction is based on the previous 24 hourly temperatures. The values of temperature were scaled between 0 and 1 with MinMaxScaler before training to improve the stability of the training of the neural network. To prevent overfitting, early stopping was implemented in the training process.

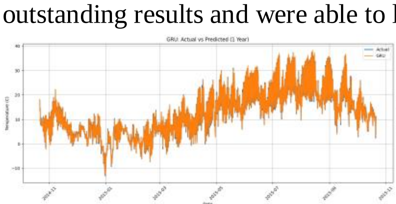
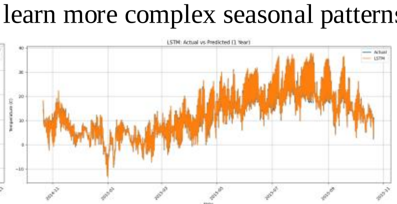
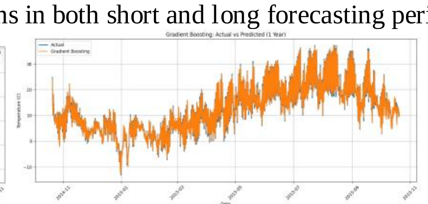
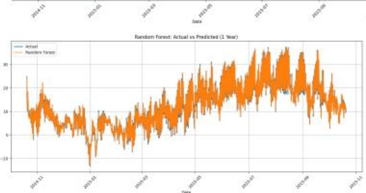
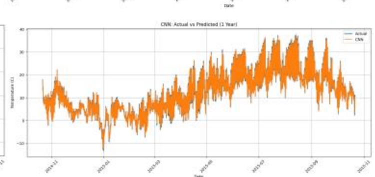
7. Experimental Setup

A training set and a testing set were created without altering the temporal order of the data set. No shuffling was used as it is important in time series forecasting to retain chronological order.

The performance of the model was assessed using: RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), sMAPE (Symmetric Mean Absolute Percentage Error), R^2 Score. In addition, the ARIMA models were validated. The rolling forecasting validation was also performed for the ARIMA models to mimic realistic forecasting situations.

8. Results

The findings indicated that this showed significant differences between the traditional statistical models and the modern machine learning/deep learning models. The Gradient Boosting method had the lowest error, RMSE, and the highest R2 score among all the methods in the overall forecasting accuracy. The model did not do a good job of adapting to the strong nonlinear seasonal structure of the data set, which led to a poor performance for SVR. It was predicting a fairly level average rather than actual temperature changes. The deep learning models have shown outstanding results and were able to learn more complex seasonal patterns in both short and long forecasting periods




.



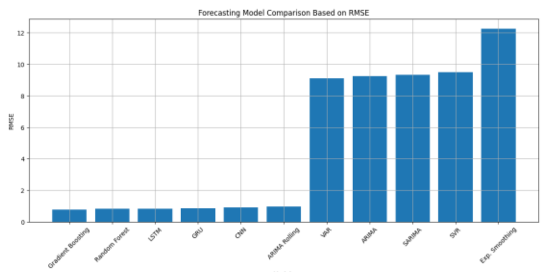
9. Discussion

The results show the significant impact of model complexity on the forecasting accuracy of weather data. Traditional statistical frameworks like ARIMA and Exponential Smoothing could only pick up some of the seasonal patterns, and have limited performance capabilities at longer forecasting periods. They became more and more convergent in predicting more and more average values, but they were not very good at capturing the rapid fluctuations.

The use of engineered lag variables and rolling statistics in the machine learning models resulted in a substantial improvement in the forecasting performance, as these models were able to learn nonlinear relationships within the data. Gradient Boosting and Random Forest were especially effective at following local temperature changes.

Deep learning models yielded some of the most realistic predictions since they learned temporal dependences directly from the sequential data. The LSTM and GRU networks proved to be very successful in modeling long-term dependencies and periodic patterns.

All models performed better than each other, with Gradient Boosting slightly better in terms of the metrics, while the LSTM and GRU models yielded visually smoother forecasts on longer time horizons. Overall, project illustrates the benefits of using machine learning and deep learning techniques for predicting highly seasonal weather data.



10. Conclusion

This project was a study of several forecasting methods for hourly temperature forecasts based on existing weather data. Exploratory analysis showed that the data had strong daily and yearly seasonality and nonlinear fluctuations. The conventional statistical models provided poor prediction power in forecasting the nonlinear long-term dynamics. The machine learning and deep learning models greatly enhanced forecasting accuracy. Gradient Boosting had the highest overall performance and an RMSE of 0.770 with an R2 of 0.993. The other models, LSTM, GRU, CNN and Random Forest, were also found to have a very good forecasting accuracy. The project was successful in the sense that it reached its goal to predict the temperature and to compare the temperature forecasting methods of various model families.

11. Future Work

There are several improvements that could be made to increase the forecasting accuracy:

- The usage of multivariate forecasting for humidity, pressure and wind variables.
- Doing bigger hyperparameter optimization.
- Applying Transformer-based deep learning architectures
- Testing probabilistic forecasts techniques.
- Incorporating external climate indicators
- Using ensemble forecasting techniques

Future research could also investigate hybrid models that combine statistical and deep learning approaches.

12. References

- Hyndman, R. J., & Athanasopoulos, G. *Forecasting: Principles and Practice*, Chapters 5, 7–9, https://robjhyndman.com/uwafiles/fpp-notes.pdf

- Box, G. E. P., Jenkins, G. M., & Reinsel, G. *Time Series Analysis*, Chapters 2–5 and 9, *https://repo.darmajaya.ac.id/4781/1/Time%20Series%20Analysis_%20Forecasting%20and%20Control%20%28%20PDFDrive%20%29.pdf*

- TensorFlow Documentation, https://www.tensorflow.org/api_docs

- Scikit-learn Documentation, https://scikit-learn.org/stable/

- Statsmodels Documentation, https://www.statsmodels.org/stable/index.html

- Jena Climate Dataset / Weather Dataset Source, https://www.kaggle.com/datasets/muthuj7/weather-dataset

- Python Libraries:
  - pandas
  - numpy
  - matplotlib
  - scikit-learn
  - statsmodels
  - tensorflow